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1  import pandas as pd
2  from sklearn.preprocessing import LabelEncoder
3  from sklearn.tree import DecisionTreeClassifier
4  from sklearn.model_selection import train_test_split
5  data = {
6      "age": [22, 35, 26, 40, 28, 50],
7      "income": [25000, 50000, 32000, 62000, 42000, 80000],
8      "browsing_duration": [5, 15, 7, 20, 10, 25],
9      "device_type": ["mobile", "desktop", "mobile", "tablet", "desktop", "mobile"],
10     "purchase": ["no", "yes", "no", "yes", "yes", "yes"]
11 }
12 df = pd.DataFrame(data)
13 le_device = LabelEncoder()
14 le_purchase = LabelEncoder()
15 df["device_type"] = le_device.fit_transform(df["device_type"])
16 df["purchase"] = le_purchase.fit_transform(df["purchase"])
17 X = df[["age", "income", "browsing_duration", "device_type"]]
18 y = df["purchase"]
19 X_train, X_test, y_train, y_test = train_test_split(*arrays: X,y,test_size=0.2)
20 model = DecisionTreeClassifier()
21 model.fit(X_train,y_train)
22 new_customer = pd.DataFrame( data: [[30,45000,12,1]],
23 columns=["age","income","browsing_duration","device_type"])
24 prediction = model.predict(new_customer)
25 print("Purchase Prediction:", le_purchase.inverse_transform(prediction))

```

Purchase Prediction: ['yes']